

MATH3881/5881: Statistical Machine Learning Theory

Week 2: Tutorial Solutions — Theory of Feedforward Neural Networks

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Worked solutions to the Week 2 tutorial problems. Problem numbers and parts match the question sheet.

Problem 1: Why we need hidden layers, XOR

- (a) *Proof by contradiction.* Suppose there exists $(w_1, w_2, b) \in \mathbb{R}^3$ correctly labelling all four XOR points. Then the four labelling constraints are:

$$(0, 0) \mapsto 0 : \quad b \leq 0, \quad (2.1)$$

$$(0, 1) \mapsto 1 : \quad w_2 + b > 0, \quad (2.2)$$

$$(1, 0) \mapsto 1 : \quad w_1 + b > 0, \quad (2.3)$$

$$(1, 1) \mapsto 0 : \quad w_1 + w_2 + b \leq 0. \quad (2.4)$$

Step 1. (Add (2.2) and (2.3).)

$$w_1 + w_2 + 2b > 0. \quad (\star)$$

Step 2. (Rewrite (2.4).)

$$-(w_1 + w_2 + b) \geq 0. \quad (\star\star)$$

Step 3. (Add $(\star)$ and $(\star\star)$.)

$$b > 0.$$

This contradicts (2.1). Hence no linear classifier exists.

Geometric picture. The two “1” points $(0, 1)$ and $(1, 0)$ lie on the line $x_1 + x_2 = 1$; the two “0” points $(0, 0)$ and $(1, 1)$ lie at the opposite corners. No single straight line can put both “1”s on one side and both “0”s on the other.

- (b) For each of the four binary inputs, compute the pre-activation $z^{[1]} = W^{[1]}x + b^{[1]}$, the hidden activation $a^{[1]} = \sigma_{\text{ReLU}}(z^{[1]})$, and the output $f_{\theta}(x) = W^{[2]}a^{[1]} + b^{[2]}$:

x	$z^{[1]} = (x_1 - x_2, x_2 - x_1)^{\top}$	$a^{[1]} = \sigma_{\text{ReLU}}(z^{[1]})$	$f_{\theta}(x) = a_1^{[1]} + a_2^{[1]}$
$(0, 0)$	$(0, 0)$	$(0, 0)$	0
$(0, 1)$	$(-1, 1)$	$(0, 1)$	1
$(1, 0)$	$(1, -1)$	$(1, 0)$	1
$(1, 1)$	$(0, 0)$	$(0, 0)$	0

The network outputs the XOR label in every case.

Problem 2: Why the UAT works for ReLU networks

- (a) For each $i \in \{1, \dots, N\}$, set $x_i^* := -b_i/w_i$.

Step 1. (Single-term structure.) Without loss of generality $w_i > 0$ (the case $w_i < 0$ is symmetric). Then $w_i x + b_i \leq 0$ iff $x \leq x_i^*$, so

$$\alpha_i \sigma_{\text{ReLU}}(w_i x + b_i) = \begin{cases} 0, & x \leq x_i^*, \\ \alpha_i (w_i x + b_i), & x \geq x_i^*. \end{cases}$$

This is piecewise linear in x with exactly one slope change, located at $x = x_i^*$.

Step 2. (Slope-change points of the sum.) A sum of functions can change slope only at a point where at least one summand changes slope. Therefore the slope of f_θ can change only at the (at most) N points $x_1^*, \dots, x_N^*$.

Step 3. (Counting pieces.) These (at most) N slope-change points partition $\mathbb{R}$ into at most $N + 1$ open intervals. On any such interval, every summand $\alpha_i \sigma_{\text{ReLU}}(w_i x + b_i)$ is linear in x , and so is the constant c . Hence f_θ is linear on each interval, that is, f_θ is piecewise linear with at most $N + 1$ pieces.

- (b) Let $f^* : [a, b] \rightarrow \mathbb{R}$ be continuous.

Step 1. (Fix an accuracy.) Let $\epsilon > 0$.

Step 2. (Approximate f^* by a piecewise-linear function.) By the result quoted in the question, there is a piecewise-linear function $g : [a, b] \rightarrow \mathbb{R}$ with

$$\sup_{x \in [a, b]} |g(x) - f^*(x)| < \epsilon.$$

Suppose g has M linear pieces, with slope-change points

$$\xi_1 < \xi_2 < \dots < \xi_{M-1} \quad \text{inside } [a, b].$$

Step 3. (Realise g exactly as a width- N ReLU network with $N = M - 1$.) Choose the parameters as follows:

- For each i , choose (w_i, b_i) so that $-b_i/w_i = \xi_i$, placing the i -th slope change of f_θ at ξ_i .
- Choose each α_i so that the slope change of f_θ at ξ_i matches the slope change of g at ξ_i .
- Choose c so that $f_\theta(a) = g(a)$.

With these parameters, f_θ and g agree on $[a, b]$ since they are piecewise linear with the same break points, the same slope changes, and the same value at the left endpoint.

Step 4. (Conclude.) Combining Steps 2 and 3,

$$\sup_{x \in [a, b]} |f^*(x) - f_\theta(x)| = \sup_{x \in [a, b]} |f^*(x) - g(x)| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, for each desired accuracy ϵ we can choose N large enough (namely $N = M(\epsilon) - 1$, where $M(\epsilon)$ is the number of pieces needed for the piecewise-linear approximation in Step 2) that some one-hidden-layer ReLU network with N units approximates f^* uniformly on $[a, b]$ to within ϵ . This is the univariate Universal Approximation Theorem for ReLU networks.